

Case Study Part Two

[illegible]

The new NPV is higher than the original NPV because the discount rate used for the project decreased from 12% to approximately 8%. A lower discount rate increases the present value of the future cash flow, resulting in a higher NPV. This reflects the fact that using a company specific WACC provides a more accurate measure of the firm's true cost of capital. Because my company's WACC is lower than the assumed 12% the project appears more valuable when evaluated using the correct discount rate.

Two assumptions are implicitly made when applying the WACC method:

1. The project has the same risk as the firm's existing operations.

WACC is only valid when the project's market risk matches the average risk of the company's current investments. Referencing back to lecture note 11, the WACC method applies when a new project is effectively a "carbon copy" of the firm's existing business. For my project, this assumption is likely to be held because the project is a scale expansion of the company's current business and operations, and uses the same revenue model, margins, and working capital cycle as the rest of the firm.

2. The firm maintains a constant debt to equity ratio.

The WACC formula assumes the company will keep the same capital structure over the life of the project, meaning the weights of the debt and equity remain stable. Referencing back to the lecture notes again, this is a key assumption for the WACC method. In practice, firms do not maintain leverage perfectly, and sometimes, given the circumstances, they will rearrange the capital components. For my project, though, this assumption is reasonable because the project scale is small relative to the firm value and has a short time horizon for the project's life cycle. If the project had a longer life cycle and halfway through the Fed became extremely dovish, making lending and debt financing more attractive, the firm might decide to allocate more debt effectively by changing the capital structure in the middle of the project.

Risk Free Rate

Given that the project's expected life cycle is 5 years it pairs well with the 5YR U.S Treasury yields which represent the risk-free rate. We use Treasuries as a risk-free rate comparison because we know the risk of default is extremely low, or non-existent, being as they are backed by the U.S government. Using the selected firm's beta, which represents its volatility or risk relative to the market, and the market's expected return based off historical data, we can then calculate the required rate of return for taking on the extra risk. This is referred to as the Capital Asset Pricing Model in its simplest form.

Cost of Debt Estimation

A company's true cost of debt is the return the market requires for the bond given the risk of default, which is the expected return. Investors know that since corporate bonds are only backed by the reputation and financial security that the firm represents, there will always be a nonzero chance of default on coupon payments or redemption at maturity. The firm promises a fixed income, and its rate of return expressed as yield to maturity (YTM) is an overestimate because there will always be risk of defaulting to varying degrees.

Equity Beta

Equity beta measures systematic risk, and it's the firm's level of market risk that cannot be diversified away. A beta of 1 would mean that the firm moves directly in tandem with the market. If the market contracts, so does the firm; the market experiences growth, and so would the firm. Conversely, a negative equity beta would mean that it moves in the opposite direction of the

market, and a 0 would mean there is no systematic relationship. The equity beta represents how volatile and in what direction the firm moves either with, against, or exhibits no relation to market movements. This equity beta is then used in the CAPM model to compute the firm's cost of equity.

Corporate Tax Rate

With Microsoft's (MSFT) capital structure being almost entirely equity financed, the 28% tax rate has only a marginal effect on its interest rate tax shield. Debt makes up roughly 3% of the firm's capital, so increasing the corporate tax rate from 21% to 28% leads to only a very small change in WACC, from 7.96381% to 7.96184%, a difference of .00197.

However, the tax rate has a much larger impact on the projects Net Present Value. The NPV falls from \$18.544 billion at a 21% tax rate to \$16.577 billion at a 28% tax rate, a decline of approximately \$1.967 billion. This occurs because the higher tax rate directly reduces Microsoft's unlevered net income, which feeds into free cash flow. Since Microsoft uses very little debt, there is no substantial interest tax shield to offset the higher tax burden. As a result, the reduction in after tax operating income flows through the cash flows and significantly lowers the project's net present value.